TRAINER KEY. DO NOT DISTRIBUTE. Day 2, Session 2: What changes at the cutoff? (regression discontinuity)

Source: H5-A. Local jump -721 AED (95% interval -943 to -498; 545 businesses, scores 56-60); five-point window -784 (-934 to -634; 1,260 businesses). As a reduction: 721, interval 498 to 943: both below the 1,000 AED rule. Manager age jumps at 58 (about -5.3 years); no score-setting diagnostic is supplied; the result applies at the cut-off, not to typical participants or the whole city.

Q1 card answers (group-specific; costs in AED):

Card	Sum at/below	Sum above	Mean at/below	Mean above	Raw gap
A	5,800	10,990	966.7	1,831.7	-865
B	9,100	13,630	1,516.7	2,271.7	-755
C	8,630	14,980	1,438.3	2,496.7	-1,058
D	5,910	11,640	985.0	1,940.0	-955
E	5,740	15,490	956.7	2,581.7	-1,625
F	8,630	10,560	1,438.3	1,760.0	-322
G	5,850	13,880	975.0	2,313.3	-1,338
H	10,140	8,620	1,690.0	1,436.7	+253

Raw gap = mean at/below minus mean above. Use the row for the group's own card; the worksheet blanks in Q1 are shown here as the example Card A.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H5-A: The cut-off study summary

GreenWaste is a fictional training case. Use this study summary and the figures on page 2 as your source for today's questions.

§1 | Who and what determined participation?

GreenWaste offers equipment subsidies, installation help and training. The city file has **10,000 businesses**: 4,794 participant businesses and 5,206 comparison businesses (businesses that did not receive GreenWaste). Businesses with efficiency scores **58 or below participated**; those above 58 did not. Higher scores mean more efficient. The running variable is the score; **58 is the cut-off**. This rule is followed in the file, but it is not a lottery.

§2 | What does the fitted result compare?

The local model uses **545 businesses within two points of 58 (scores 56-60)**. The outcome is annual waste cost in AED per business, measured **12 months after**. Separate straight cost/score slopes are fitted on either side. The coefficient compares the fitted costs approaching 58 from the participating side versus the non-participating side. It is not the average gap across the whole city, or the small raw mean gap on a card.

Result row	Coefficient AED	95% interval AED	Businesses
At/below cut-off: local jump	-721	-943 to -498	545

Cost 12 months after GreenWaste ~ below * distance; distance = score - 58. Separate linear slopes; window 56-60; HC2 robust interval. Reference: fitted cost above the cut-off at 58. The coefficient compares the fitted limits at the cut-off.

§3 | Which checks are supplied or missing?

A five-point window (scores 53-63) uses **1,260 businesses** and estimates **-784 AED**, interval **-934 to -634**. This sensitivity comparison does not prove causal validity. **Pre-programme manager age also jumps at 58: about -5.3 years in the five-point window.** It raises a concern about who is compared and other explanations for a cost jump. It does not prove a particular bias. No score-setting or manipulation diagnostic is supplied. The

evaluator should investigate the scoring process, who could influence scores and other discontinuities. A causal reading requires comparable businesses immediately either side apart from programme access, without relevant sorting or other important jumps.

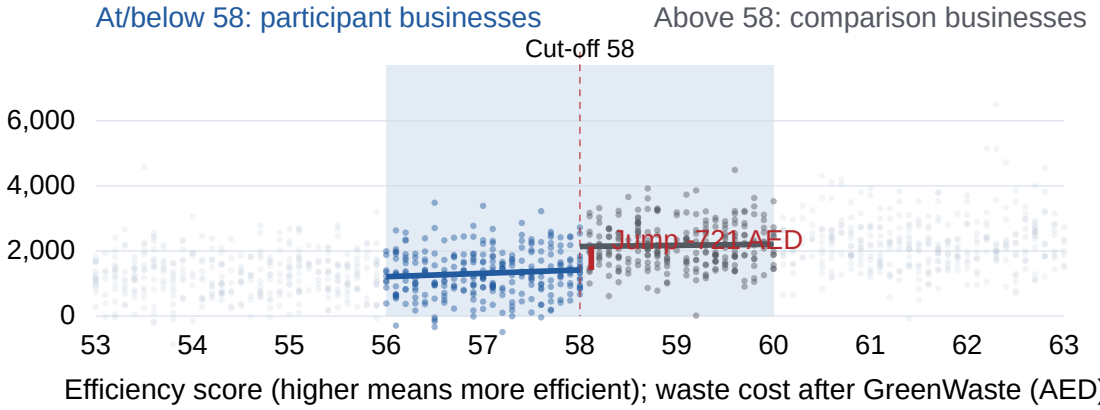
§4 | What follows for the decision?

On a positive-reduction scale, the local estimate is **721 AED**, interval **498-943**. Both are below the fictional **1,000 AED annual-saving minimum**, conditional on the model. Precision does not resolve the design concern. Most city participants are outside this local window. A result at the cut-off is not an estimated average effect for typical participants or the whole city. Full programme costs and national effects are not supplied.

Source: GreenWaste city data (10,000 businesses). Figures show records from the GreenWaste data and fitted lines, except the sorting illustration, which is hypothetical.

Page 2 | H5-A: visual reference

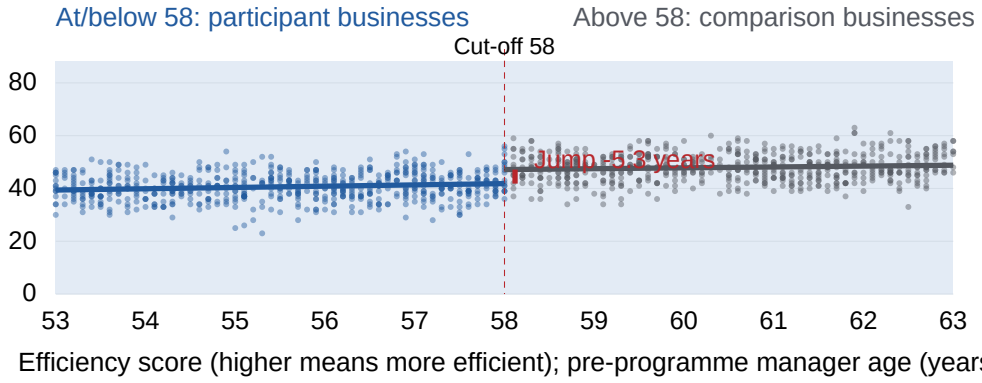
The fitted jump at 58: two-point window



Dots are city businesses in the GreenWaste data with scores 53-63. Shading marks scores 56-60 used in the main model. Lines approach the cut-off; the red gap is between their fitted limits.

Window	Businesses	Jump AED	95% interval AED
Within 2 points	545	-721	-943 to -498
Within 5 points	1,260	-784	-934 to -634

A pre-programme characteristic: manager age



This age discontinuity is a credibility concern, not a GreenWaste effect on age. Optional extension: adding age to the two-point model gives **-632 AED**, HC2 interval **-857 to -407**. Adjustment alone does not establish a valid local comparison.

Page 3 | Q1-Q2: arithmetic and interpretation

Q1 | Record your group's raw card gap

Card letter: **A** . Use your **separate A-H card**, six businesses per side. Costs are rounded to ten AED. Scores are displayed to one decimal, but each business is placed on its side using its exact score.

Calculation	At/below 58	Above 58
Sum of six costs	**5,800**	**10,990**
Mean: sum divided by 6	**966.7**	**1,831.7**

At/below mean minus above mean: **-865** AED.

Place the letter and gap in the **raw-gap lane** on the number line. Why might other groups get different gaps?

> Each group has only six businesses per side, chosen differently, and costs vary a lot between businesses; a small sample's raw gap moves with which businesses are on the card. Gaps range from +253 to -1,625.

Q2 | Read the fitted local jump

Use H5-A §2/table (page 1) and the fitted plot on page 2. Label the gap at the cut-off. Why does the fitted estimate not equal your card gap?

> The fitted estimate uses all 545 businesses in the window with separate slopes and compares the fitted limits exactly at 58; the card gap compares raw means of six businesses that sit at different scores. The card gap is not the fitted jump or an established effect.

Interpret the coefficient: comparison, sign, units, timing and local population. Name what must be credible before calling it an effect.

> Fitted annual cost 12 months after GreenWaste for businesses just at/below 58 is 721 AED lower than for businesses just above 58 (negative = lower cost), among businesses near the cut-off. A causal reading needs comparable businesses immediately either side apart from programme access: no sorting/manipulation of scores and no other important jumps at 58.

Convert the negative interval to positive reduction. Where are the estimate and interval relative to 1,000 AED? What remains unresolved?

> Reduction = 721 AED, interval 498 to 943 AED. The estimate and the whole interval are below 1,000 AED, so it does not meet the rule. Unresolved: whether the local comparison is valid (age jump, score setting) and whether the result applies beyond the cut-off.

Page 4 | Q3: what would make this credible?

Use H5-A §1-4 (page 1) and page 2's figures. Mark **supported, concern or unknown**, citing evidence and a reason.

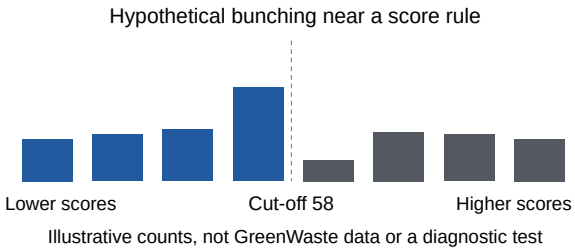
Check	Judgement, source and reason
Score rule followed	**Supported (businesses with scores of 58 or below took part; those above did not; this rule is followed in the file)**
Local comparison/window disclosed	**Supported/disclosed (545 businesses, scores 56-60, with a 5-point check of 1,260)**
Other pre-programme differences	**Concern (manager age jumps by about -5.3 years at 58)**
Score setting/manipulation investigated	**Unknown (no score-setting or manipulation diagnostic is supplied)**
Whole-city effect established	**Not established (the estimate is local to the cut-off; most participants are outside the window)**

What changes when the window widens? What does stability still not prove?

> The estimate changes only a little (-784, interval -934 to -634), so it is not sensitive to the window. Stability does not prove validity: a wider window includes less comparable businesses and does not address the age jump or score setting.

Why does the age jump matter? What would you ask the evaluator to investigate?

> Manager age is fixed before the programme, so a jump at 58 shows the groups differ in something other than GreenWaste (a concern for comparability, not an effect on age). Ask: how were scores set and by whom, could businesses influence them, what other characteristics jump at 58, and what does the score distribution near 58 look like?



This is hypothetical, not a supplied manipulation test. Ask who set scores, when and whether businesses could influence which side they landed on.

Your independent reading and request

Keep H5-A open; close AI responses and peer comments. Write one supported local-result interpretation and one specific request. Explain how the answer could change your advice.

> Example: "Near the cut-off, costs 12 months after are about 721 AED lower for participants, but the age difference and unknown score-setting mean I cannot call it a programme effect, and it does not meet 1,000 AED. I would ask who set the scores and whether businesses could influence them. If scores were manipulated, my advice would change to hold off." Accept any supported reading plus a specific request.

Page 5 | AI Snapshot: local result, wider claim

This response was written for the exercise. It is not a real AI answer. The prompt was: **Explain without jargon who this result applies to, and what a policy adviser should check before expansion.** Do not run this first prompt.

> The score threshold provides a regression discontinuity comparison. The fitted annual cost immediately below the cut-off is 721 AED lower than immediately above it, with a robust 95% interval of 498 to 943 AED lower. A wider five-point window also gives lower fitted costs, which is a useful sensitivity comparison.
>
> ~~Businesses on either side of the threshold are effectively interchangeable because their scores are close.~~ ~~The manager-age difference is a background characteristic and does not affect this conclusion; adjusting for it would remove any remaining selection bias.~~ ~~The stable direction across windows confirms that the threshold isolates the programme effect.~~
>
> Implementation costs and practical capacity should still be checked. ~~However, because most participants have scores below 58, this result describes the typical participant and supports citywide expansion.~~ ~~The interval is precise enough to show that the 1,000 AED saving requirement is met.~~

Underline support and cross out overclaims. Check **H5-A, pages 1-2**. Does a sensible caveat repair an unsupported assurance?

Improved prompt: run and compare

Explain H5-A without jargon to a policy adviser considering expansion. Businesses with a score at or below 58 took part; the local linear model uses scores 56-60 and 545 businesses, with separate slopes. The jump is -721 AED in annual waste cost 12 months after GreenWaste, HC2 interval -943 to -498; as a reduction this is 721, interval 498-943. Most city participants are outside this local window. Pre-programme manager age jumps by about -5.3 years in the five-point window. No score-setting/manipulation diagnostic is supplied. The fictional annual-saving minimum is 1,000 AED.

Ask essential clarifying questions first. Explain who the local comparison describes, what the coefficient and interval show, and what evidence is needed for a causal reading. Distinguish rule compliance from comparison validity. Request specific missing evidence; do not dismiss the age check, promise that adjustment fixes it, generalise citywide or claim that the saving minimum is met. Cite H5-A.

Copy this prompt and H5-A into **Microsoft Copilot**. Keep your AI good-practice card beside you. Page 6 is the prepared paper alternative.

Which claim improved? What still needs checking against the source?

> Look for: coefficient/interval/window comparison kept; the claims that proximity proves interchangeability, that age is irrelevant, that stability confirms causality, that the result describes the typical participant and that 1,000 is met removed or qualified; specific missing evidence named. Still to check: every number, the age jump and the rule.

Page 6 | Prepared response: paper alternative

Use this for the same task if Copilot is unavailable. This response was written for the exercise. It is not a real AI answer, and it is not additional study evidence.

> Before recommending expansion, I would ask how scores were assigned and recorded, whether businesses could influence them, and why manager age differs at the same threshold. I would also ask for other pre-programme checks and reasons for the chosen local window.
>
> H5-A compares fitted annual waste management costs 12 months after GreenWaste, immediately either side of score 58, using 545 businesses in scores 56-60 and separate slopes. It reports 721 AED lower cost on the participating side, with a model-based robust interval of 498 to 943 AED lower. A programme-effect reading requires a credible local comparison without other important jumps or sorting at the cut-off. The age discontinuity is a concern, not proof of a particular bias; controlling for age alone does not establish validity.
>
> A similar wider-window result is useful but does not settle these concerns. This is evidence near the cut-off, not an estimated average effect for all city participants. Both the estimate and interval are below the fictional 1,000 AED minimum, conditional on the model. I would investigate the design concern before relying on a causal interpretation, and request evidence for businesses further from the cut-off before extrapolating.

Check even a careful response

Match its comparison and numbers to H5-A §2 and page 2. Match design concerns and reach to §3-4. Does it distinguish recorded evidence, missing checks and an assumption?

> Yes: recorded evidence (721, 498 to 943, 545 businesses, 5-point window), missing checks (scoring process, age jump) and the assumption (comparable businesses at the cut-off). Numbers match §2 and page 2.

Which assurance from page 5 has been removed? Is any wording still too strong or unclear?

> Removed: that businesses either side are "effectively interchangeable", that age "does not affect" the conclusion or that adjusting "would remove bias", that stability "confirms" the effect, that it describes the typical participant, and that the 1,000 requirement is met. Wording to tighten: "would investigate" is fine; "model-based" could be explained. Accept sensible edits.

Return to page 4 for your **own** interpretation and request, with AI responses closed.